```
import torch
import torch.nn as nn
import torch.optim as optim
from torchvision import datasets, transforms, models
from torch.utils.data import DataLoader

# ========================
# 1. CONFIG & DEVICE
# ========================
batch_size = 128
epochs = 30
learning_rate = 0.01

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print(f"Training on: {device}")

# For reproducibility (optional)
torch.manual_seed(42)
if device.type == "cuda":
    torch.cuda.manual_seed_all(42)

# ========================
# 2. DATASET & DATALOADERS
# ========================

# FashionMNIST: 28x28 grayscale images, 10 classes
# We'll use simple augmentation on train set to help generalization
train_transform = transforms.Compose([
    transforms.RandomHorizontalFlip(p=0.5),
    transforms.RandomCrop(28, padding=2),
    transforms.ToTensor(),
    transforms.Normalize((0.5,), (0.5,))  # grayscale mean/std
])

test_transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize((0.5,), (0.5,))
])

data_root = "~/.pytorch/F_MNIST_data/"

print("Loading FashionMNIST...")
train_dataset = datasets.FashionMNIST(
    root=data_root,
    train=True,
    download=True,
    transform=train_transform
)

test_dataset = datasets.FashionMNIST(
    root=data_root,
    train=False,
    download=True,
    transform=test_transform
)

train_loader = DataLoader(train_dataset, batch_size=batch_size, shuffle=True, num_workers=2, pin_memory=
test_loader = DataLoader(test_dataset, batch_size=batch_size, shuffle=False, num_workers=2, pin_memory=T

class_names = [
    "T-shirt/top",
    "Trouser",
    "Pullover",
    "Dress",
    "Coat",
    "Sandal",
    "Shirt",
    "Sneaker",
    "Bag",
    "Ankle boot"
]

# ------------------------
```

```python
# ==========================
# 3. MODEL DEFINITION
# ==========================

def get_fashion_resnet(num_classes: int = 10):
    """
    ResNet-18 adapted for FashionMNIST (1-channel, 28x28 images).
    """
    model = models.resnet18(weights=None)  # start from scratch for simplicity

    # Change input conv layer to accept 1-channel (grayscale)
    model.conv1 = nn.Conv2d(
        in_channels=1,
        out_channels=64,
        kernel_size=3,
        stride=1,
        padding=1,
        bias=False
    )

    # Remove the large initial maxpool — images are small (28x28)
    model.maxpool = nn.Identity()

    # Change the final fully connected layer to output num_classes logits
    model.fc = nn.Linear(in_features=512, out_features=num_classes)

    return model

model = get_fashion_resnet(num_classes=len(class_names))
model = model.to(device)

# ==========================
# 4. LOSS, OPTIMIZER, SCHEDULER
# ==========================
criterion = nn.CrossEntropyLoss(label_smoothing=0.1)

optimizer = optim.AdamW(
    model.parameters(),
    lr=learning_rate,
    weight_decay=1e-4
)

scheduler = optim.lr_scheduler.CosineAnnealingLR(
    optimizer,
    T_max=epochs
)
```

```
Training on: cuda
Loading FashionMNIST...
```

```python
# ==========================
# 5. TRAIN & EVAL FUNCTIONS
# ==========================

def train_one_epoch(model, loader, optimizer, criterion, device):
    model.train()
    running_loss = 0.0
    total = 0
    correct = 0

    for images, labels in loader:
        images = images.to(device)
        labels = labels.to(device)

        optimizer.zero_grad()
        outputs = model(images)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()

        running_loss += loss.item() * images.size(0)

        _, preds = torch.max(outputs, dim=1)
        correct += (preds == labels).sum().item()
```

```
        total += labels.size(0)

    epoch_loss = running_loss / total
    epoch_acc = 100.0 * correct / total
    return epoch_loss, epoch_acc


def evaluate(model, loader, criterion, device):
    model.eval()
    running_loss = 0.0
    total = 0
    correct = 0

    with torch.no_grad():
        for images, labels in loader:
            images = images.to(device)
            labels = labels.to(device)

            outputs = model(images)
            loss = criterion(outputs, labels)

            running_loss += loss.item() * images.size(0)

            _, preds = torch.max(outputs, dim=1)
            correct += (preds == labels).sum().item()
            total += labels.size(0)

    epoch_loss = running_loss / total
    epoch_acc = 100.0 * correct / total
    return epoch_loss, epoch_acc
```

```
# =========================
# 6. TRAINING LOOP
# =========================

best_acc = 0.0
best_model_path = "fashion_resnet_best.pth"

print(f"Starting classification training for {epochs} epochs...")

for epoch in range(epochs):
    train_loss, train_acc = train_one_epoch(model, train_loader, optimizer, criterion, device)
    val_loss, val_acc = evaluate(model, test_loader, criterion, device)

    scheduler.step()
    current_lr = optimizer.param_groups[0]['lr']

    if val_acc > best_acc:
        best_acc = val_acc
        torch.save(model.state_dict(), best_model_path)

    print(
        f"Epoch [{epoch+1}/{epochs}] "
        f"| LR: {current_lr:.5f} "
        f"| Train Loss: {train_loss:.4f}, Train Acc: {train_acc:.2f}% "
        f"| Val Loss: {val_loss:.4f}, Val Acc: {val_acc:.2f}% "
        f"(Best Val Acc: {best_acc:.2f}%)"
    )

print("\nTraining complete.")
print(f"Best validation accuracy: {best_acc:.2f}%")
print(f"Best model saved to: {best_model_path}")
```

```
Starting classification training for 30 epochs...
Epoch [1/30] | LR: 0.00997 | Train Loss: 1.0932, Train Acc: 74.07% | Val Loss: 1.6411, Val Acc: 48.00% (
Epoch [2/30] | LR: 0.00989 | Train Loss: 0.8044, Train Acc: 86.43% | Val Loss: 0.8067, Val Acc: 86.11% (
Epoch [3/30] | LR: 0.00976 | Train Loss: 0.7471, Train Acc: 89.13% | Val Loss: 0.7385, Val Acc: 88.98% (
Epoch [4/30] | LR: 0.00957 | Train Loss: 0.7168, Train Acc: 90.29% | Val Loss: 0.7492, Val Acc: 89.40% (
Epoch [5/30] | LR: 0.00933 | Train Loss: 0.6998, Train Acc: 91.12% | Val Loss: 0.6988, Val Acc: 90.68% (
Epoch [6/30] | LR: 0.00905 | Train Loss: 0.6889, Train Acc: 91.61% | Val Loss: 0.7029, Val Acc: 90.62% (
Epoch [7/30] | LR: 0.00872 | Train Loss: 0.6739, Train Acc: 92.30% | Val Loss: 0.6854, Val Acc: 91.62% (
Epoch [8/30] | LR: 0.00835 | Train Loss: 0.6640, Train Acc: 92.67% | Val Loss: 0.6777, Val Acc: 92.17% (
Epoch [9/30] | LR: 0.00794 | Train Loss: 0.6558, Train Acc: 93.11% | Val Loss: 0.6718, Val Acc: 92.04% (
Epoch [10/30] | LR: 0.00750 | Train Loss: 0.6502, Train Acc: 93.29% | Val Loss: 0.6675, Val Acc: 92.44%
```

```
Epoch [11/30] | LR: 0.00703 | Train Loss: 0.6398, Train Acc: 93.78% | Val Loss: 0.6537, Val Acc: 93.19%
Epoch [12/30] | LR: 0.00655 | Train Loss: 0.6348, Train Acc: 94.01% | Val Loss: 0.6496, Val Acc: 93.32%
Epoch [13/30] | LR: 0.00604 | Train Loss: 0.6263, Train Acc: 94.45% | Val Loss: 0.6472, Val Acc: 93.43%
Epoch [14/30] | LR: 0.00552 | Train Loss: 0.6186, Train Acc: 94.75% | Val Loss: 0.6763, Val Acc: 92.23%
Epoch [15/30] | LR: 0.00500 | Train Loss: 0.6142, Train Acc: 94.90% | Val Loss: 0.6404, Val Acc: 93.86%
Epoch [16/30] | LR: 0.00448 | Train Loss: 0.6053, Train Acc: 95.33% | Val Loss: 0.6481, Val Acc: 93.46%
Epoch [17/30] | LR: 0.00396 | Train Loss: 0.5996, Train Acc: 95.57% | Val Loss: 0.6288, Val Acc: 94.31%
Epoch [18/30] | LR: 0.00345 | Train Loss: 0.5922, Train Acc: 95.93% | Val Loss: 0.6365, Val Acc: 94.12%
Epoch [19/30] | LR: 0.00297 | Train Loss: 0.5867, Train Acc: 96.19% | Val Loss: 0.6255, Val Acc: 94.60%
Epoch [20/30] | LR: 0.00250 | Train Loss: 0.5795, Train Acc: 96.55% | Val Loss: 0.6262, Val Acc: 94.36%
Epoch [21/30] | LR: 0.00206 | Train Loss: 0.5728, Train Acc: 96.80% | Val Loss: 0.6288, Val Acc: 94.58%
Epoch [22/30] | LR: 0.00165 | Train Loss: 0.5667, Train Acc: 97.08% | Val Loss: 0.6243, Val Acc: 94.76%
Epoch [23/30] | LR: 0.00128 | Train Loss: 0.5606, Train Acc: 97.39% | Val Loss: 0.6216, Val Acc: 95.04%
Epoch [24/30] | LR: 0.00095 | Train Loss: 0.5565, Train Acc: 97.59% | Val Loss: 0.6227, Val Acc: 94.93%
Epoch [25/30] | LR: 0.00067 | Train Loss: 0.5504, Train Acc: 97.90% | Val Loss: 0.6206, Val Acc: 95.22%
Epoch [26/30] | LR: 0.00043 | Train Loss: 0.5471, Train Acc: 98.02% | Val Loss: 0.6201, Val Acc: 95.08%
Epoch [27/30] | LR: 0.00024 | Train Loss: 0.5430, Train Acc: 98.27% | Val Loss: 0.6199, Val Acc: 95.17%
Epoch [28/30] | LR: 0.00011 | Train Loss: 0.5408, Train Acc: 98.37% | Val Loss: 0.6200, Val Acc: 95.25%
Epoch [29/30] | LR: 0.00003 | Train Loss: 0.5385, Train Acc: 98.53% | Val Loss: 0.6206, Val Acc: 95.20%
Epoch [30/30] | LR: 0.00000 | Train Loss: 0.5363, Train Acc: 98.67% | Val Loss: 0.6202, Val Acc: 95.27%

Training complete.
Best validation accuracy: 95.27%
Best model saved to: fashion_resnet_best.pth
```

```
#takes 25 mins to run ^
```

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Price Prediction

```
import pandas as pd
import numpy as np
from pathlib import Path

from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.ensemble import HistGradientBoostingRegressor
from sklearn.pipeline import Pipeline
```

```
DATA_PATH = Path("/content/transactions_articles_sample.csv")

df = pd.read_csv(DATA_PATH)

print("Original shape:", df.shape)
print(df.head().T)  # transpose to inspect columns quickly
```

```
colour_group_name                                                Dark Blue
```

```
colour_group_name                                                  Dark Blue
perceived_colour_value_id                                                   2
perceived_colour_value_name                                       Medium Dusty
perceived_colour_master_id                                                  2
perceived_colour_master_name                                             Blue
department_no                                                            8558
department_name                                                Men Sport Tops
index_code                                                                  S
index_name                                                             Sport
index_group_no                                                             26
index_group_name                                                       Sport
section_no                                                                 22
section_name                                                    Men H&M Sport
garment_group_no                                                        1007
garment_group_name                                                    Outdoor
detail_desc                   Jersey sports top with a zip at the top and st...

                                                                            4
t_dat                                                             2018-09-28
customer_id                   e86cfe736ab46f06691768e7a87ede453bca1b523531d7...
article_id                                                         621381011
price                                                               0.033881
sales_channel_id                                                           2
product_code                                                          621381
prod_name                                                  Jade Denim Trash TRS
product_type_no                                                         272
product_type_name                                                    Trousers
product_group_name                                       Garment Lower body
graphical_appearance_no                                               1010016
graphical_appearance_name                                               Solid
colour_group_code                                                           9
colour_group_name                                                       Black
perceived_colour_value_id                                                   4
perceived_colour_value_name                                              Dark
perceived_colour_master_id                                                  5
perceived_colour_master_name                                            Black
department_no                                                            1747
department_name                                                      Trousers
index_code                                                                  D
index_name                                                           Divided
index_group_no                                                              2
index_group_name                                                     Divided
section_no                                                                 53
section_name                                                Divided Collection
garment_group_no                                                        1009
garment_group_name                                                   Trousers
detail_desc                   High-waisted jeans in washed superstretch deni...
```

```
df["price_real_sek"] = df["price"] * 10000
df["price_usd"] = df["price_real_sek"] / 10.5   # approx SEK→USD conversion

print(df[["price", "price_real_sek", "price_usd"]].head())
```

```
      price  price_real_sek  price_usd
0  0.013542      135.423729  12.897498
1  0.011847      118.474576  11.283293
2  0.025407      254.067797  24.196933
3  0.032186      321.864407  30.653753
4  0.033881      338.813559  32.267958
```

```
cols_to_drop = [
    "t_dat", "customer_id", "article_id", "product_code",
    "prod_name", "detail_desc", "price", "price_real_sek"  # we now only use price_usd
]

df = df.drop(columns=[c for c in cols_to_drop if c in df.columns])

print("After dropping unnecessary columns:", df.shape)

# Drop rows missing our target
df = df.dropna(subset=["price_usd"])

# Optional: downsample for RAM
MAX_ROWS = 500_000
if len(df) > MAX_ROWS:
    df = df.sample(MAX_ROWS, random_state=42).reset_index(drop=True)
```

```
After dropping unnecessary columns: (2000000, 23)
```

```
target_col = "price_usd"
y = df[target_col]
X = df.drop(columns=[target_col])

categorical_cols = [c for c in X.columns if X[c].dtype == "object"]
numeric_cols = [c for c in X.columns if c not in categorical_cols]

print("Categorical cols:", categorical_cols)
print("Numeric cols:", numeric_cols)
```

```
Categorical cols: ['product_type_name', 'product_group_name', 'graphical_appearance_name', 'colour_group
Numeric cols: ['sales_channel_id', 'product_type_no', 'graphical_appearance_no', 'colour_group_code', 'p
```

```
#preprocessing
preprocessor = ColumnTransformer(
    transformers=[
        ("cat", OneHotEncoder(handle_unknown="ignore", sparse_output=False), categorical_cols),
        ("num", "passthrough", numeric_cols)
    ]
)

regressor = HistGradientBoostingRegressor(
    max_depth=8,
    learning_rate=0.1,
    max_iter=200,
    min_samples_leaf=50,
    random_state=42
)

model = Pipeline(steps=[
    ("preprocess", preprocessor),
    ("regressor", regressor)
])
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
# ========================
# 7. TRAIN MODEL
# ========================
model.fit(X_train, y_train)

# ========================
# 8. EVALUATE
# ========================
y_pred = model.predict(X_test)

rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)

print(f"RMSE (USD): {rmse:.2f}")
print(f"R² Score:   {r2:.4f}")
```

```
RMSE (USD): 10.96
R² Score:   0.7499
```

```
import joblib

joblib.dump(model, "hm_price_regressor.pkl")
print("Saved regressor to hm_price_regressor.pkl")
```

```
Saved regressor to hm_price_regressor.pkl
```

```
# Assuming df is the same dataframe used for training the regressor
price_by_type = df.groupby("product_type_name")["price_usd"].median().sort_values()
print(price_by_type)
```

```
product_type_name
Hair string           3.890234
```

```
Hair clip             4.019370
Earrings              4.826473
Chem. cosmetics       4.826473
Hair ties             4.826473
                       ...
Weekend/Gym bag      54.866828
Boots                56.481033
Jacket               64.552058
Outdoor Waistcoat    64.552058
Coat                 96.836158
Name: price_usd, Length: 101, dtype: float64
```

```
# price_by_type you already have:
price_by_type = df.groupby("product_type_name")["price_usd"].median().sort_values()

print("Global median:", df["price_usd"].median())
print(price_by_type.tail(10))  # just to inspect the most expensive ones
```

```
Global median: 24.196933010492284
product_type_name
Underdress           48.410008
Blazer               48.410008
Outdoor trousers     48.410008
Outdoor overall      50.831316
Giftbox              53.656174
Weekend/Gym bag      54.866828
Boots                56.481033
Jacket               64.552058
Outdoor Waistcoat    64.552058
Coat                 96.836158
Name: price_usd, dtype: float64
```

```
def median_for_keywords(keywords):
    """
    keywords: list of strings like ["t-shirt", "top"]
    Returns median price_usd over all rows whose product_type_name
    contains ANY of those keywords (case-insensitive).
    Falls back to global median if nothing matches.
    """
    mask = False
    for kw in keywords:
        mask = mask | df["product_type_name"].str.contains(kw, case=False, na=False)
    if mask.sum() == 0:
        return float(df["price_usd"].median())
    return float(df.loc[mask, "price_usd"].median())
```

```
class_names = [
    "T-shirt/top",
    "Trouser",
    "Pullover",
    "Dress",
    "Coat",
    "Sandal",
    "Shirt",
    "Sneaker",
    "Bag",
    "Ankle boot"
]

class_to_median_price = {
    "T-shirt/top": median_for_keywords(["t-shirt", "top", "tee"]),
    "Trouser":     median_for_keywords(["trouser", "pants", "jeans"]),
    "Pullover":    median_for_keywords(["pullover", "sweater", "knit"]),
    "Dress":       median_for_keywords(["dress"]),
    "Coat":        median_for_keywords(["coat", "jacket", "waistcoat"]),
    "Sandal":      median_for_keywords(["sandal"]),
    "Shirt":       median_for_keywords(["shirt", "blouse"]),
    "Sneaker":     median_for_keywords(["sneaker", "trainer", "shoe"]),
    "Bag":         median_for_keywords(["bag", "backpack", "pouch"]),
    "Ankle boot":  median_for_keywords(["boot"])
}

print("Class → median price (USD):")
```

```
for k, v in class_to_median_price.items():
    print(f"{k:12s}: ${v:.2f}")
```

```
Class → median price (USD):
T-shirt/top : $12.90
Trouser     : $32.27
Pullover    : $25.81
Dress       : $32.27
Coat        : $64.55
Sandal      : $19.35
Shirt       : $19.35
Sneaker     : $32.27
Bag         : $24.20
Ankle boot  : $56.48
```

```
def estimate_price_from_image(image_path: str):
    _, pred_class_name, probs = predict_class_from_image(image_path)
    base_price = class_to_median_price.get(pred_class_name, df["price_usd"].median())

    print(f"Predicted clothing class: {pred_class_name}")
    print(f"Estimated price (median for this class): ${base_price:.2f}")
    return base_price
```

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```
import torch
import torch.nn as nn
import torch.optim as optim
from torchvision import datasets, transforms, models
from torch.utils.data import DataLoader
from PIL import Image
import numpy as np

# ========================
# 1. CONFIG & DEVICE
# ========================
batch_size = 128
epochs = 30
learning_rate = 0.01
weight_decay = 1e-4  # FIX: This was missing!

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print(f"Training on: {device}")

torch.manual_seed(42)
if device.type == "cuda":
    torch.cuda.manual_seed_all(42)

# ========================
# 2. DATASET & DATALOADERS
# ========================

train_transform = transforms.Compose([
    transforms.RandomHorizontalFlip(p=0.5),
    transforms.RandomCrop(28, padding=2),
    transforms.ToTensor(),
    transforms.Normalize((0.5,), (0.5,))
```

```
])

test_transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize((0.5,), (0.5,))
])

data_root = "~/.pytorch/F_MNIST_data/"

print("Loading FashionMNIST...")
train_dataset = datasets.FashionMNIST(
    root=data_root,
    train=True,
    download=True,
    transform=train_transform
)

test_dataset = datasets.FashionMNIST(
    root=data_root,
    train=False,
    download=True,
    transform=test_transform
)

train_loader = DataLoader(train_dataset, batch_size=batch_size, shuffle=True, num_workers=2, pin_memory
test_loader = DataLoader(test_dataset, batch_size=batch_size, shuffle=False, num_workers=2, pin_memory=

class_names = [
    "T-shirt/top",
    "Trouser",
    "Pullover",
    "Dress",
    "Coat",
    "Sandal",
    "Shirt",
    "Sneaker",
    "Bag",
    "Ankle boot"
]

# ==========================
# 3. MODEL DEFINITION
# ==========================

def get_fashion_resnet(num_classes: int = 10):
    """
    For FashionMNIST (grayscale): train from scratch with 1-channel input
    """
    model = models.resnet18(weights=None)
    model.conv1 = nn.Conv2d(
        in_channels=1,
        out_channels=64,
        kernel_size=3,
        stride=1,
        padding=1,
        bias=False
    )
    model.maxpool = nn.Identity()
    model.fc = nn.Linear(in_features=512, out_features=num_classes)
    return model

def get_pretrained_fashion_resnet(num_classes: int = 10):
    """
    For real RGB images: use pre-trained weights and fine-tune
    """
    model = models.resnet18(weights='IMAGENET1K_V1')  # Pre-trained on RGB

    # Keep the RGB input (3 channels) - don't modify conv1
    # Only replace the final layer
    model.fc = nn.Linear(in_features=512, out_features=num_classes)

    return model

# Use the grayscale version for FashionMNIST training
```

```python
model = get_fashion_resnet(num_classes=len(class_names))
model = model.to(device)

# ========================
# 4. LOSS, OPTIMIZER, SCHEDULER
# ========================
criterion = nn.CrossEntropyLoss(label_smoothing=0.1)

optimizer = optim.AdamW(
    model.parameters(),
    lr=learning_rate,
    weight_decay=weight_decay  # Now defined
)

scheduler = optim.lr_scheduler.CosineAnnealingLR(
    optimizer,
    T_max=epochs
)

# ========================
# 5. TRAIN & EVAL FUNCTIONS
# ========================

def train_one_epoch(model, loader, optimizer, criterion, device):
    model.train()
    running_loss = 0.0
    total = 0
    correct = 0

    for images, labels in loader:
        images = images.to(device)
        labels = labels.to(device)

        optimizer.zero_grad()
        outputs = model(images)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()

        running_loss += loss.item() * images.size(0)

        _, preds = torch.max(outputs, dim=1)
        correct += (preds == labels).sum().item()
        total += labels.size(0)

    epoch_loss = running_loss / total
    epoch_acc = 100.0 * correct / total
    return epoch_loss, epoch_acc


def evaluate(model, loader, criterion, device):
    model.eval()
    running_loss = 0.0
    total = 0
    correct = 0

    with torch.no_grad():
        for images, labels in loader:
            images = images.to(device)
            labels = labels.to(device)

            outputs = model(images)
            loss = criterion(outputs, labels)

            running_loss += loss.item() * images.size(0)

            _, preds = torch.max(outputs, dim=1)
            correct += (preds == labels).sum().item()
            total += labels.size(0)

    epoch_loss = running_loss / total
    epoch_acc = 100.0 * correct / total
    return epoch_loss, epoch_acc
```

```python
# ==========================
# 6. TRAINING LOOP
# ==========================

best_acc = 0.0
best_model_path = "fashion_resnet_best.pth"

print(f"Starting classification training for {epochs} epochs...")

for epoch in range(epochs):
    train_loss, train_acc = train_one_epoch(model, train_loader, optimizer, criterion, device)
    val_loss, val_acc = evaluate(model, test_loader, criterion, device)

    scheduler.step()
    current_lr = optimizer.param_groups[0]['lr']

    if val_acc > best_acc:
        best_acc = val_acc
        torch.save(model.state_dict(), best_model_path)

    print(
        f"Epoch [{epoch+1}/{epochs}] "
        f"| LR: {current_lr:.5f} "
        f"| Train Loss: {train_loss:.4f}, Train Acc: {train_acc:.2f}% "
        f"| Val Loss: {val_loss:.4f}, Val Acc: {val_acc:.2f}% "
        f"(Best Val Acc: {best_acc:.2f}%)"
    )

print("\nTraining complete.")
print(f"Best validation accuracy: {best_acc:.2f}%")
print(f"Best model saved to: {best_model_path}")
```

```
Training on: cuda
Loading FashionMNIST...
100%|██████████| 26.4M/26.4M [00:02<00:00, 10.8MB/s]
100%|██████████| 29.5k/29.5k [00:00<00:00, 192kB/s]
100%|██████████| 4.42M/4.42M [00:01<00:00, 3.50MB/s]
100%|██████████| 5.15k/5.15k [00:00<00:00, 15.5MB/s]
Starting classification training for 30 epochs...
Epoch [1/30] | LR: 0.00997 | Train Loss: 1.0858, Train Acc: 74.38% | Val Loss: 2.3735, Val Acc: 35.10% (
Epoch [2/30] | LR: 0.00989 | Train Loss: 0.7977, Train Acc: 86.80% | Val Loss: 0.7838, Val Acc: 86.67% (
Epoch [3/30] | LR: 0.00976 | Train Loss: 0.7439, Train Acc: 89.27% | Val Loss: 0.7668, Val Acc: 88.19% (
Epoch [4/30] | LR: 0.00957 | Train Loss: 0.7173, Train Acc: 90.28% | Val Loss: 0.7185, Val Acc: 90.54% (
Epoch [5/30] | LR: 0.00933 | Train Loss: 0.6994, Train Acc: 91.17% | Val Loss: 0.6873, Val Acc: 91.31% (
Epoch [6/30] | LR: 0.00905 | Train Loss: 0.6893, Train Acc: 91.60% | Val Loss: 0.6921, Val Acc: 91.15% (
Epoch [7/30] | LR: 0.00872 | Train Loss: 0.6738, Train Acc: 92.32% | Val Loss: 0.6859, Val Acc: 91.65% (
Epoch [8/30] | LR: 0.00835 | Train Loss: 0.6644, Train Acc: 92.72% | Val Loss: 0.6962, Val Acc: 91.21% (
Epoch [9/30] | LR: 0.00794 | Train Loss: 0.6554, Train Acc: 93.04% | Val Loss: 0.6781, Val Acc: 91.72% (
Epoch [10/30] | LR: 0.00750 | Train Loss: 0.6504, Train Acc: 93.31% | Val Loss: 0.6849, Val Acc: 91.65%
Epoch [11/30] | LR: 0.00703 | Train Loss: 0.6421, Train Acc: 93.67% | Val Loss: 0.6549, Val Acc: 93.08%
Epoch [12/30] | LR: 0.00655 | Train Loss: 0.6343, Train Acc: 94.03% | Val Loss: 0.6501, Val Acc: 93.38%
Epoch [13/30] | LR: 0.00604 | Train Loss: 0.6257, Train Acc: 94.42% | Val Loss: 0.6474, Val Acc: 93.11%
Epoch [14/30] | LR: 0.00552 | Train Loss: 0.6185, Train Acc: 94.77% | Val Loss: 0.6608, Val Acc: 92.83%
Epoch [15/30] | LR: 0.00500 | Train Loss: 0.6143, Train Acc: 94.90% | Val Loss: 0.6425, Val Acc: 93.56%
Epoch [16/30] | LR: 0.00448 | Train Loss: 0.6055, Train Acc: 95.35% | Val Loss: 0.6395, Val Acc: 93.57%
Epoch [17/30] | LR: 0.00396 | Train Loss: 0.6004, Train Acc: 95.57% | Val Loss: 0.6280, Val Acc: 93.95%
Epoch [18/30] | LR: 0.00345 | Train Loss: 0.5925, Train Acc: 95.92% | Val Loss: 0.6296, Val Acc: 94.26%
Epoch [19/30] | LR: 0.00297 | Train Loss: 0.5871, Train Acc: 96.20% | Val Loss: 0.6311, Val Acc: 94.26%
Epoch [20/30] | LR: 0.00250 | Train Loss: 0.5789, Train Acc: 96.50% | Val Loss: 0.6323, Val Acc: 94.22%
Epoch [21/30] | LR: 0.00206 | Train Loss: 0.5726, Train Acc: 96.78% | Val Loss: 0.6242, Val Acc: 94.56%
Epoch [22/30] | LR: 0.00165 | Train Loss: 0.5668, Train Acc: 97.10% | Val Loss: 0.6245, Val Acc: 94.83%
Epoch [23/30] | LR: 0.00128 | Train Loss: 0.5610, Train Acc: 97.38% | Val Loss: 0.6243, Val Acc: 94.79%
Epoch [24/30] | LR: 0.00095 | Train Loss: 0.5564, Train Acc: 97.58% | Val Loss: 0.6230, Val Acc: 94.84%
Epoch [25/30] | LR: 0.00067 | Train Loss: 0.5508, Train Acc: 97.84% | Val Loss: 0.6194, Val Acc: 95.11%
Epoch [26/30] | LR: 0.00043 | Train Loss: 0.5457, Train Acc: 98.14% | Val Loss: 0.6225, Val Acc: 94.93%
Epoch [27/30] | LR: 0.00024 | Train Loss: 0.5440, Train Acc: 98.19% | Val Loss: 0.6218, Val Acc: 95.13%
Epoch [28/30] | LR: 0.00011 | Train Loss: 0.5407, Train Acc: 98.38% | Val Loss: 0.6217, Val Acc: 95.17%
Epoch [29/30] | LR: 0.00003 | Train Loss: 0.5381, Train Acc: 98.51% | Val Loss: 0.6221, Val Acc: 95.15%
Epoch [30/30] | LR: 0.00000 | Train Loss: 0.5372, Train Acc: 98.53% | Val Loss: 0.6219, Val Acc: 95.13%

Training complete.
Best validation accuracy: 95.17%
Best model saved to: fashion_resnet_best.pth
```

```python
!pip install git+https://github.com/openai/CLIP.git
```

```python
import torch
import clip
from PIL import Image

# 1. LOAD CLIP MODEL

device = "cuda" if torch.cuda.is_available() else "cpu"
model, preprocess = clip.load("ViT-B/32", device=device)

print(f"Using device: {device}")
print("CLIP model loaded successfully!")

# 2. DEFINE CATEGORIES

class_names = [
    "T-shirt",
    "Trouser",
    "Pullover",
    "Dress",
    "Coat",
    "Sandal",
    "Shirt",
    "Sneaker",
    "Bag",
    "Ankle boot"
]

class_to_median_price = {
    "T-shirt": 12.90,
    "Trouser": 32.27,
    "Pullover": 25.81,
    "Dress": 32.27,
    "Coat": 64.55,
    "Sandal": 19.35,
    "Shirt": 19.35,
    "Sneaker": 32.27,
    "Bag": 24.20,
    "Ankle boot": 56.48,
}

# Create text prompts (better descriptions improve accuracy)
text_prompts = [
    "a photo of a t-shirt",
    "a photo of trousers",
    "a photo of a pullover sweater",
    "a photo of a dress",
    "a photo of a coat",
    "a photo of sandals",
    "a photo of a shirt",
    "a photo of sneakers",
    "a photo of a bag",
    "a photo of ankle boots"
]

# Encode text prompts
text_tokens = clip.tokenize(text_prompts).to(device)
with torch.no_grad():
    text_features = model.encode_text(text_tokens)
    text_features /= text_features.norm(dim=-1, keepdim=True)

# 3. PREDICTION FUNCTION

def predict_clothing_clip(image_path: str):
    """
    Predict clothing class using CLIP (zero-shot)
    Works on real images without any training!
    """
    # Load and preprocess image
    image = preprocess(Image.open(image_path)).unsqueeze(0).to(device)

    # Get image features
    with torch.no_grad():
        image_features = model.encode_image(image)
        image_features /= image_features.norm(dim=-1, keepdim=True)
```

```
        # Calculate similarity scores
        similarity = (100.0 * image_features @ text_features.T).softmax(dim=-1)
        values, indices = similarity[0].topk(5)

    # Get prediction
    pred_idx = indices[0].item()
    pred_class = class_names[pred_idx]
    confidence = values[0].item()

    # Get price
    price = class_to_median_price.get(pred_class, 24.20)

    # Display results
    print(f"\n{'='*60}")
    print(f"Image: {image_path}")
    print(f"Model: CLIP (Zero-Shot, No Training)")
    print(f"{'='*60}")
    print(f"Predicted: {pred_class.upper()}")
    print(f"Confidence: {confidence*100:.2f}%")
    print(f"\nTop-5 predictions:")
    for i in range(5):
        idx = indices[i].item()
        prob = values[i].item()
        print(f"  {class_names[idx]:15s}: {prob*100:.2f}%")
    print(f"\nEstimated price: ${price:.2f} USD")
    print(f"{'='*60}\n")

    return pred_class, price

# 4. USAGE

# Test on your images - NO TRAINING NEEDED!
predict_clothing_clip("/content/coat.png")
predict_clothing_clip("/content/shirt.png")
predict_clothing_clip("/content/ldress.jpg")

# This should give MUCH better results on real photos!
```

```
============================================================
Image: /content/ldress.jpg
Model: CLIP (Zero-Shot, No Training)
============================================================
Predicted: DRESS
Confidence: 99.27%

Top-5 predictions:
  Dress          : 99.27%
  Shirt          : 0.36%
  Coat           : 0.24%
  T-shirt        : 0.04%
  Bag            : 0.03%

Estimated price: $32.27 USD
============================================================

('Dress', 32.27)
```

Start coding or generate with AI.

```
!pip install scikit-learn
```

```
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: numpy>=1.19.5 in /usr/local/lib/python3.12/dist-packages (from scikit-lea
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-lear
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-lea
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from sci
```

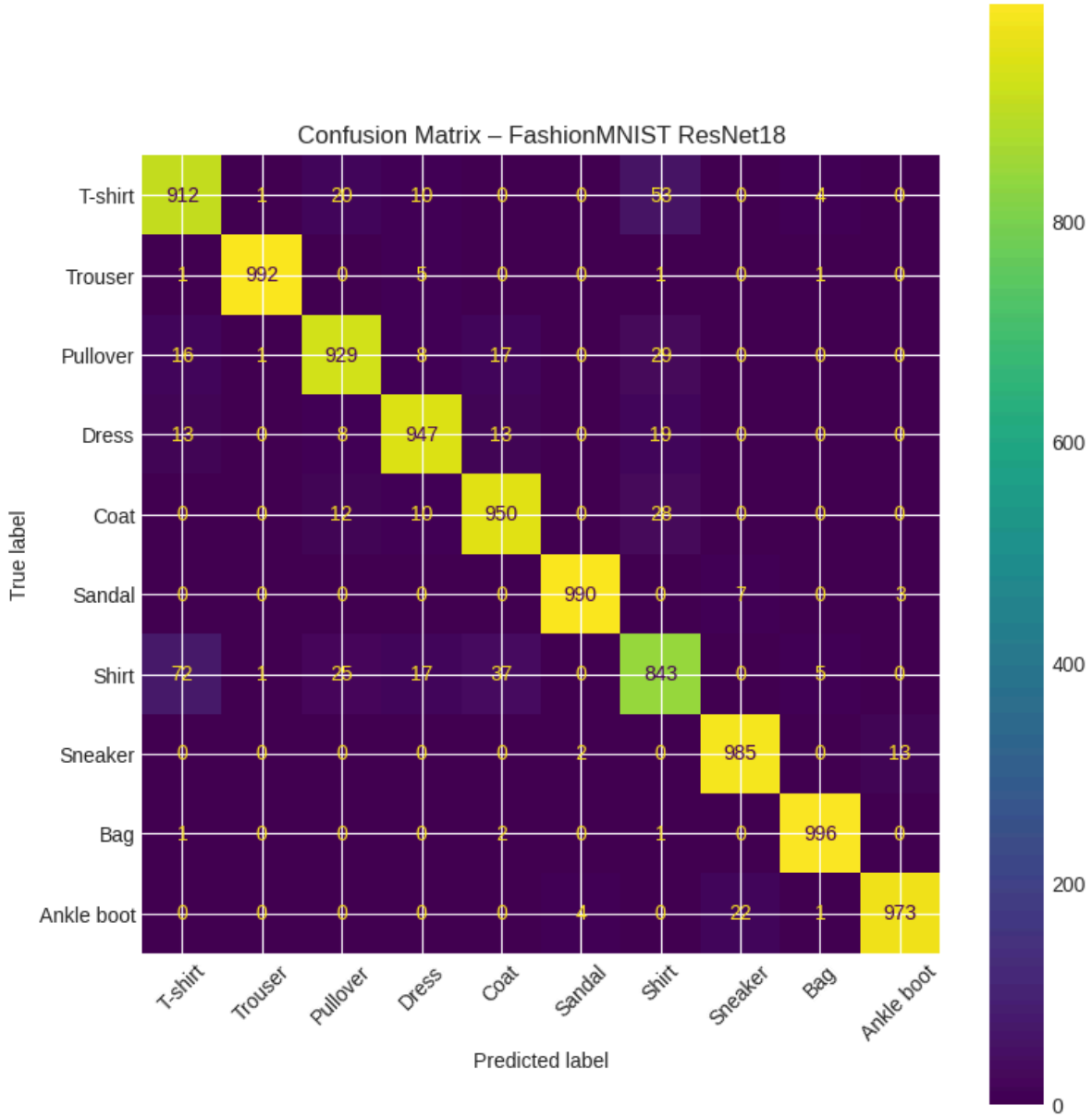
```
import torch
import numpy as np
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

# Recreate the same model architecture used for training
fashion_model = get_fashion_resnet(num_classes=len(class_names))
fashion_model.load_state_dict(torch.load("fashion_resnet_best.pth", map_location=device))
fashion_model.to(device)
fashion_model.eval()
```

```
      (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (downsample): Sequential(
        (0): Conv2d(64, 128, kernel_size=(1, 1), stride=(2, 2), bias=False)
        (1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      )
    )
    (1): BasicBlock(
```

```
  )
  (layer4): Sequential(
    (0): BasicBlock(
      (conv1): Conv2d(256, 512, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), bias=False)
      (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (relu): ReLU(inplace=True)
      (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (downsample): Sequential(
        (0): Conv2d(256, 512, kernel_size=(1, 1), stride=(2, 2), bias=False)
        (1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      )
    )
    (1): BasicBlock(
      (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (relu): ReLU(inplace=True)
      (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    )
  )
  (avgpool): AdaptiveAvgPool2d(output_size=(1, 1))
  (fc): Linear(in_features=512, out_features=10, bias=True)
)
```

```
all_labels = []
all_preds = []

with torch.no_grad():
    for images, labels in test_loader:
        images = images.to(device)
        labels = labels.to(device)

        outputs = fashion_model(images)
        _, preds = torch.max(outputs, dim=1)

        all_labels.append(labels.cpu().numpy())
        all_preds.append(preds.cpu().numpy())

all_labels = np.concatenate(all_labels)
all_preds = np.concatenate(all_preds)
```

```
cm = confusion_matrix(all_labels, all_preds)

fig, ax = plt.subplots(figsize=(8, 8))
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=class_names)
disp.plot(ax=ax, xticks_rotation=45)
plt.title("Confusion Matrix — FashionMNIST ResNet18")
plt.tight_layout()
plt.show()
```



Start coding or generate with AI.

Start coding or generate with AI.

```
import matplotlib.pyplot as plt

# 1. Reconstruct history from your epoch prints

train_losses = [
    1.0858,
    0.7977,
    0.7439,
    0.7173,
    0.6994,
    0.6893,
    0.6738,
    0.6644,
    0.6554,
    0.6504,
    0.6421,
    0.6343,
    0.6257,
    0.6185,
    0.6143,
    0.6055,
    0.6004,
    0.5925,
    0.5871,
    0.5789,
    0.5726,
```

```
    0.5668,
    0.5610,
    0.5564,
    0.5508,
    0.5457,
    0.5440,
    0.5407,
    0.5381,
    0.5372,
]

val_losses = [
    2.3735,
    0.7838,
    0.7668,
    0.7185,
    0.6873,
    0.6921,
    0.6859,
    0.6962,
    0.6781,
    0.6849,
    0.6549,
    0.6501,
    0.6474,
    0.6608,
    0.6425,
    0.6395,
    0.6280,
    0.6296,
    0.6311,
    0.6323,
    0.6242,
    0.6245,
    0.6243,
    0.6230,
    0.6194,
    0.6225,
    0.6218,
    0.6217,
    0.6221,
    0.6219,
]

train_accuracies = [
    74.38,
    86.80,
    89.27,
    90.28,
    91.17,
    91.60,
    92.32,
    92.72,
    93.04,
    93.31,
    93.67,
    94.03,
    94.42,
    94.77,
    94.90,
    95.35,
    95.57,
    95.92,
    96.20,
    96.50,
    96.78,
    97.10,
    97.38,
    97.58,
    97.84,
    98.14,
    98.19,
    98.38,
    98.51,
    98.53,
```

```
]

val_accuracies = [
    35.10,
    86.67,
    88.19,
    90.54,
    91.31,
    91.15,
    91.65,
    91.21,
    91.72,
    91.65,
    93.08,
    93.38,
    93.11,
    92.83,
    93.56,
    93.57,
    93.95,
    94.26,
    94.26,
    94.22,
    94.56,
    94.83,
    94.79,
    94.84,
    95.11,
    94.93,
    95.13,
    95.17,
    95.15,
    95.13,
]

epochs_range = range(1, len(train_losses) + 1)

# 2. Loss curve
plt.figure()
plt.plot(epochs_range, train_losses, label="Train Loss")
plt.plot(epochs_range, val_losses, label="Val Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("Training vs Validation Loss (ResNet18 on FashionMNIST)")
plt.legend()
plt.grid(True)
plt.show()

# 3. Accuracy curve
plt.figure()
plt.plot(epochs_range, train_accuracies, label="Train Accuracy")
plt.plot(epochs_range, val_accuracies, label="Val Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy (%)")
plt.title("Training vs Validation Accuracy (ResNet18 on FashionMNIST)")
plt.legend()
plt.grid(True)
plt.show()
```

Training vs Validation Loss (ResNet18 on FashionMNIST)
Train Loss
Val Loss
Loss
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0
5
10
15
20
25
30
Epoch